Annexure-I

Data Structure and Algorithm using Java Programming

CipherSchools

A training report

Submitted in partial fulfilment of the requirements for the award of degree of

B.Tech. (Computer Science Engineering)

Submitted to

LOVELY PROFESSIONAL UNIVERSITY
PHAGWARA, PUNJAB

From June 4, 2025 to July 15, 2025

SUBMITTED BY

Name of student: Hrithik Parihar

Registration Number: 12324916

Signature of the student: *Hrithik*

Annexure-II: Student Declaration

To whom so ever it may concern

I, **Hrithik Parihar**, Registration Number **[12324916]**, hereby declare that the work done by me on **"Data Structure and Algorithm using Java Programming"** from **4 June, 2025** to **15 July, 2025**, is a record of original work for the partial fulfilment of the requirements for the award of the degree, **B.Tech. in Computer Science Engineering**.

Hrithik Parihar(12324916)

Dated: 12/08/2025

Training certificate from organization/ Company

https://www.cipherschools.com/certificate/preview?id=6885b910ca64e035786b255d

Cipher Schools

Certificate of Completion

This is to certify that

Hrithik Parihar

studying at **Lovely Professional University,** has successfully completed training in **Data Structure and Algorithm using Java Programming** from CipherSchools during the period of **June 4, 2025 to July 15, 2025** .

The training comprised 70 hours of learning , and the participant's performance has been assessed as Satisfactory.

ANURAG MISHRA

Founder CipherSchools

CipherSchools, India

Certificate ID : CSW2025-13130

Scan to verify

INTRODUCTION OF THE PROJECT UNDERTAKEN

- **Objectives of the work undertaken:** The primary objective was to gain practical, hands-on experience with Data Structures and Algorithms (DSA) using Java by designing and implementing complete problem-solving solutions. This involved understanding the theory behind various algorithms and data structures such as arrays, linked lists, stacks, queues, trees, graphs, and heaps, and integrating them seamlessly within Java programs to achieve efficiency and reliability.

- **Scope of the Work:** The scope included implementing different data structures in Java, applying suitable algorithms for various problems, and developing modular, reusable code. It also covered integrating these components into functional Java-based projects, testing for accuracy, and optimizing performance in terms of time and space complexity to ensure scalability and adaptability.

- **Importance and Applicability**: The project's importance lies in its focus on using DSA concepts in Java to develop efficient and reliable solutions. In the modern technology industry, choosing the right data structure and algorithm is essential for building optimized applications. The skills and concepts gained are directly applicable to competitive programming, technical interviews, and real-world software development.

- **Role and profile**: My role was that of a Java Programming Intern, responsible for designing and implementing problem-solving programs using DSA concepts. This included selecting appropriate data structures, writing efficient algorithms, testing solutions, and improving code performance, all under the guidance of the CipherSchools training program.

INTRODUCTION OF THE COMPANY/WORK

- **Company's Vision and Mission**: CipherSchools is a leading ed-tech platform with a mission to make high-quality, practical programming and skill development accessible to students across the globe. Their vision is to bridge the gap between academic knowledge and industry needs by offering comprehensive training in core concepts such as Data Structures and Algorithms using Java, along with other cutting-edge technologies that prepare learners for real-world challenges.

- **Origin and growth of company**: Founded to meet the rising demand for skilled developers, CipherSchools has grown rapidly in both reach and reputation. Over time, it has expanded its course offerings to include in-depth training on topics like DSA using Java, while building a strong community of learners who benefit from structured learning paths and mentorship opportunities.

- **Various departments and their functions**:As an online training platform, the company is organized into specialized teams responsible for delivering a seamless learning experience. The content creation team develops high-quality Java and DSA course materials, the technical support team assists students with programming-related issues, the mentorship team guides learners through their problem-solving journey, and the platform development team ensures a smooth and engaging online learning environment.

- **Organization chart of the company**: Being a dynamic online platform, the organization follows a structure that typically includes founders overseeing strategic decisions, content leads managing course development, and technical teams maintaining platform functionality and providing student support for learning resources like DSA in Java.

Brief description of the work done

- **Position of Internship and roles**: My position was a Summer Intern, focusing on Java programming with a specialization in Data Structures and Algorithms. My primary role was to apply theoretical knowledge by developing a project that demonstrated proficiency in implementing, optimizing, and integrating various data structures and algorithms within Java applications to solve real-world problems efficiently.

- **Activities/ equipment handled**: The main activity was the development of a problem-solving application using DSA concepts in Java. This involved creating Java classes for different data structures, implementing algorithms for efficient operations, organizing methods for modularity, and integrating these components into a functional program. The primary "equipment" included my personal computer and a development environment consisting of an IDE (such as IntelliJ IDEA or Eclipse), command-line tools, and the Java Development Kit (JDK).

- **Challenges faced and how those were tackled**: One key challenge was effectively managing the interaction between different data structures and algorithms to maintain efficiency and clarity in program flow. This was addressed by carefully structuring classes, methods, and logic, ensuring that data was passed and processed consistently. Another challenge was learning to implement complex structures like trees and graphs efficiently. This was solved by studying Java documentation and reliable online resources, which helped me properly define, construct, and manipulate these structures for optimal performance.

- **Learning outcomes**: I gained a strong practical understanding of applying Data Structures and Algorithms using Java to solve computational problems. I learned how to design and implement efficient solutions, optimize time and space complexity, and debug intricate logic errors, enhancing both my programming skills and analytical thinking.

- **Data analysis**: Data analysis was performed on the program's data structure design, such as how elements would be stored in arrays, linked lists, trees, and hash tables, to ensure efficient access, retrieval, and modification. The goal was to select the most suitable data structure for each task, ensuring scalability and performance in various problem scenarios.

CONCLUSION

The summer internship at CipherSchools was a valuable experience that provided a comprehensive understanding of Data Structures and Algorithms using Java. The training and hands-on project strengthened my theoretical knowledge and practical skills, showing how different data structures and algorithms can be integrated to create efficient, optimized, and reliable programs. The problem-solving approaches, coding practices, and analytical thinking developed during this internship are highly relevant and will serve as a strong foundation for my future career in software development.

REFERENCES

https://www.cs.cipherschools.com/courses/data-structures-and-algorithms-using-java (Accessed on July 15, 2025)